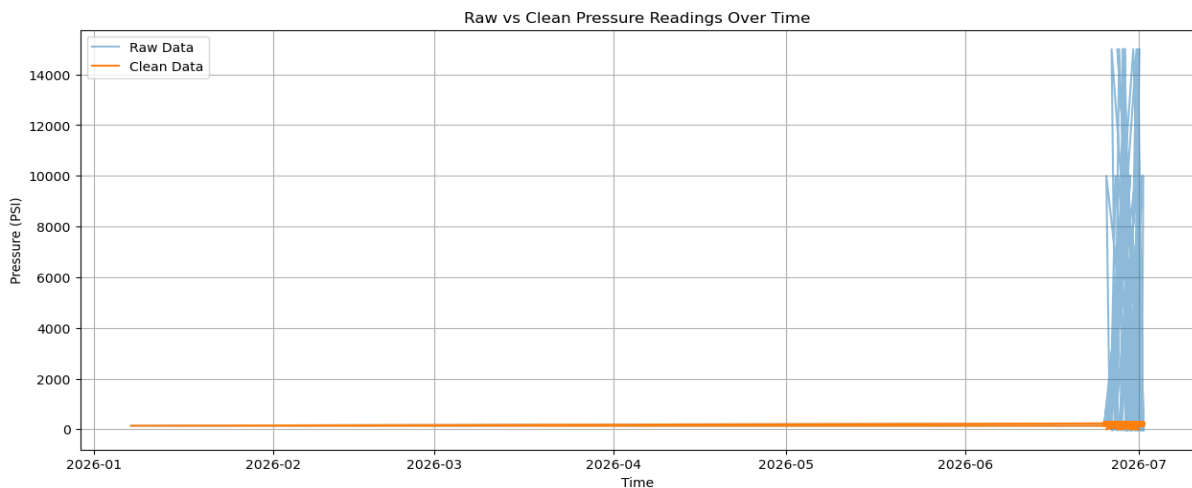
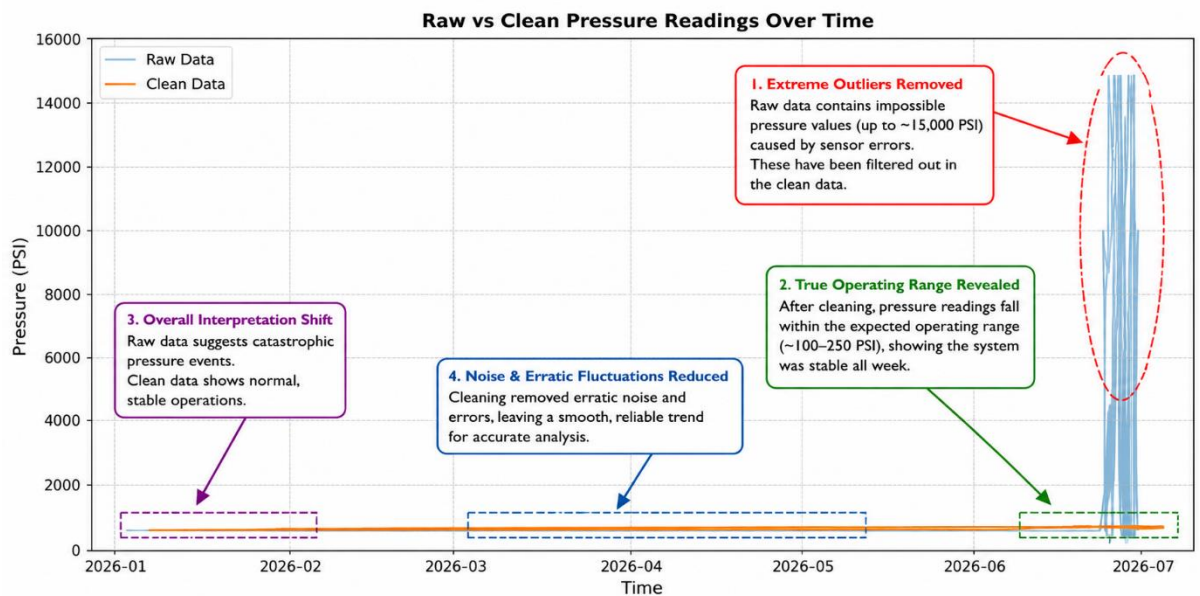


## 1. Plot generated



## 2. Visual annotation



- Context; the original sensor dataset contained missing values, duplicate observations, inconsistent categorical labels, and physically impossible sensor readings. These quality issues distorted the operational picture and increased the risk of incorrect decisions based on unreliable trends.

Insight; once the dataset was cleaned and standardized, a recurring nightly increase in the primary sensor metric became visible. The raw data suggested operational stability, but cleaning revealed a pattern that had been hidden by invalid readings and duplicated records.

Action; immediate investigation of night-shift operations is recommended. Review equipment loading, maintenance activities, and operator procedures during overnight periods. Introduce alert thresholds and enhanced monitoring to detect and respond to future spikes before they affect production performance or safety.